1. USB-C input, PTC fuse, TVS, USB ESD

J1
USB-C receptacle (USB 2.0, sink)

F1
PTC 0.75A hold, >=6V

D1
TVS 5V standoff (candidate SMAJ5.0CA)

U2
USBLC6-2SC6

R1
5.1k 1%

R2
5.1k 1%

2. 3V3 buck regulator (AP63203)

U3
AP63203WU-7 (3.3V buck)

C1
10uF 25V X7R

C2
100nF 50V

C3
100nF 25V

L1
3.9uH, Isat>=3A (candidate)

C4
22uF 10V X7R

C5
22uF 10V X7R

Power path: USB_VBUS -> F1 (PTC) -> +5V -> U3 AP63203 -> L1 -> +3V3. PWR_FLAGs mark rails fed through passives for ERC.

3. ESP32-S3-WROOM-1-N8, EN RC, RESET/BOOT

U1
ESP32-S3-WROOM-1-N8

R3
10k

R4
10k

C6
10uF 10V X7R

C7
100nF 16V

SW1
RESET

SW2
BOOT

C8
1uF 16V

4. Synthetic pulse input — 3.3V LAB SIGNAL ONLY

J2
PULSE_IN (synthetic 3.3V lab signal only)

R5
10k

R6
1k

C9
100nF (DNP)

5. Status LED

D2
Green LED

R7
1k

6. Enclosure tamper switch (advisory only)

J3
TAMPER switch

R8
10k

R9
1k

C10
100nF

7. I2C expansion and debug UART

J4
I2C expansion (3.3V)

J5
Debug UART0 (3.3V)

R10
4.7k

R11
4.7k

8. Modem daughterboard — MODEM_VIN separate, never +3V3

J7
MODEM_VIN bench supply input

J6
Modem daughterboard interface

R12
100k

R13
100k

9. Test points

TP1 TP_5V
TP2 TP_3V3
TP3 TP_GND
TP4 TP_PULSE
TP5 TP_EN
TP6 TP_BOOT
TP7 TP_VMODEM
TP8 TP_GND2
TP9 TP_VBUS

ENGINEERING DRAFT — NOT FABRICATION READY. Generated from design_v02.py.
Bench-only: current-limited low-voltage supply + synthetic 3.3V pulse source.
Never connect to a live gas meter, gas line, valve, mains or field wiring.
Candidate values from datasheet typical applications; verify before freeze.
Smart Gas Meter IoT — lab prototype

Sheet: /
File: smart_gas_meter_iot.kicad_sch

Title: Smart Gas Meter IoT — carrier board v0.2 (controlled rebuild)

Size: A3	Date: 2026-10-24	**Rev: 0.2-draft2**
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